Early Physical Reasoning and Object Permanence: Looking vs. Manual Search

Young infants demonstrate object permanence and physical reasoning in looking-time tasks months before manual cloth retrieval; search failure reflects frontostriatal inhibitory and motor planning immaturity rather than absent representation.

GRADE: MODERATE-VOE; LOW-COMPLETE-THEORY | STATUS: DRAFT | DOMAIN: LEARNING, COGNITION | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Parents frequently ask when do babies get object permanence? For half a century, the Piagetian orthodoxy held that infants do not acquire object permanence until approximately 8 to 9 months, when they first reach under a cloth to retrieve a hidden rattle, and that full mental representation remains incomplete until 18 to 24 months as evidenced by perseverative A-not-B reach errors (Piaget 1954). However, modern looking-time experiments demonstrated that young infants possess mental representations of hidden objects far earlier. Baillargeon (1987) demonstrated using rotating-drawbridge habituation displays that 3.5- to 4.5-month-old infants look significantly longer at impossible occlusion events (where a solid screen rotates 180 degrees through the space occupied by a hidden wooden box), proving that perceptual awareness of object solidity and permanence exists months before infants coordinate reaching hands. Diamond (1985) subsequently showed that the A-not-B search error does not stem from absent object permanence, but from the slow maturation of dorsolateral prefrontal cortex inhibitory control and motor working memory: infants look toward the correct hiding location B while their reaching hand perseverates to previously reinforced location A.

EVIDENCE AND CAUSAL ASSESSMENT

Experimental dissociation between looking and manual reaching. Baillargeon's violation-of-expectation (VOE) paradigm has been replicated across occlusion, containment, collision, and support configurations (Stahl & Feigenson 2015). Diamond's developmental progression across 7.5 to 12 months showed that increasing the delay between hiding and reaching produces systematic A-not-B perseveration until prefrontal inhibitory circuits mature. Search tasks and looking tasks dissociate because two-step means-end manual retrieval places heavy demands on immature motor executive coordination.

EFFECT SIZE AND PERSISTENCE

VOE looking-time differences between possible and impossible occlusion events are robust (d = 0.50 to 0.80 in tightly controlled laboratory displays). Longitudinal tracking shows that early VOE sensitivity reflects normative cognitive architecture rather than an individual predictor of school-age IQ.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a. Methodological debates center on whether prolonged staring reflects rich conceptual representation versus low-level perceptual novelty or habituation dynamics (Aslin 2007; ManyBabies 2020).

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean young babies are miniature physicists who understand abstract gravity or mechanics. Does not mean failing to reach for a hidden toy at 6 months indicates cognitive delay. Does not mean commercial flashcards or baby-physics toys accelerate brain development.

WHAT ACTION IS REASONABLY JUSTIFIED

Parents do not need special cognitive drilling toys or apps. Natural interactive games (peekaboo, hiding a toy under a blanket, watching objects roll and fall) provide ample sensory-motor scaffolding matching the infant's natural developmental timetable.

VERIFIED SOURCES & DOI IDENTIFIERS

- Baillargeon, R. (1987). Object permanence in 3.5- and 4.5-month-old infants. Developmental Psychology, 23(5), 655–664. DOI: 10.1037/0012-1649.23.5.655

- Diamond, A. (1985). Development of the ability to use recall to guide action, as indicated by infants' performance on AB. Child Development, 56(4), 868–883. DOI: 10.2307/1130099

- Stahl, A. E., & Feigenson, L. (2015). Observing the unexpected enhances infants' learning and exploration. Science, 348(6230), 91–94. DOI: 10.1126/science.aaa3799